

## The data used in the 2024 anchovy assessment

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*Data to which the 2024 assessment of the South African anchovy resource is to be conditioned are listed.*

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The data used in the South African anchovy assessment are not all raw data. Some data have already been subjected to analyses and refinements. These associated calculations are often done “behind the scenes” and their details are seldom recorded. This lack of record can result in a discontinuity in the method used to calculate data for subsequent assessments, particularly if assumptions made in the calculations are not documented and/or there is a change in personnel responsible for developing the data to be used for input to the assessment. This document serves to record the generation of the data used in the anchovy assessment carried out in 2024 – using data up to the end of 2023 - from the raw data.

### Information on Ageing

Anchovy age-length keys for the November surveys from 1992 to 1995 were derived by Prosch (De Oliveria 2003), and by Kerstan for 1990, 1992 to 1995 (unpublished data). These data are not used in the assessment.

### Anchovy Commercial Data

Monthly anchovy catch numbers are available for 1981 to 1983 (De Oliveria pers. comm.) but no LFs are available for these months. These data are not used in the assessment.

Monthly length frequencies were constructed for the anchovy landings using the method in Appendix A. Although it is possible to split the length frequencies by area from 1987, as the assessment will be run for a single stock in a single area, length frequencies for a single area only are considered. A few 15cm fish were recorded as anchovy landings in September/October 2001, but these were excluded from the assessment as this is most likely a mistake in the reported species. Anchovy catches are mostly inshore and the possibility of landing a 15cm anchovy is small.

In October and November 1984, no length frequencies were available although there were landings totalling 22 878t in October and 7 271t in November. In these cases, the length frequencies of a former month (July 1984 when 18 193t was landed) were used to estimate a raised length frequency as follows:

$$RLF_{y,missing,l} = LF_{y,previous,l} \times Tonnage_{y,missing} / Tonnage_{y,previous}$$

The LFs by month and year, i.e. the number of fish in length class  $l$  in month  $m$  of year  $y$ ,  $N_{y,l,m}$ , from 1984 to 2023 are stored in *AnchovyCatchLength.xlsx*, together with the observed tonnage in each month and year,  $ObsT_{y,m}$  (Table 1). The “raised” length frequency used in the assessment, is then calculated as follows:

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$$\hat{N}_{y,m,l} = N_{y,m,l} \times \frac{ObsT_{y,m}}{\sum_i EM_{y,l,m}}.$$

Expected mass by length class and month for October to March is calculated as:

$$EM_{y,l} = 0.007366 \times l_{mid}^{3.11124} \times N_{y,l,m}$$

and for April to September is calculated as:

$$EM_{y,l} = 0.006803 \times l_{mid}^{3.11124} \times N_{y,l,m}$$

where  $l_{mid}$  is the mid-point of the length class considered and mass is in grams and length in centimetres (de Moor 2023).

Quarterly data used in the assessments are taken over the months October  $y - 1$  to September  $y$ . For the 1984 year, there are no data for October - December 1983 and thus catch in the 4<sup>th</sup> quarter of 1983 is assumed to be zero. The October – December 2023 catches are required to fit the survey biomass in the final year.

#### Recruit catch prior to the survey

Length frequencies were also calculated from 1 April to the day before the commencement of the survey using the method in Appendix A. Inspector data (which include samples for species split) are required to do this (see Appendix A), but were not available in 1985 and 1986. For these years the daily skippers' estimates of tonnage landed were, however, available and used. Although the total tonnage landed in May 1985 and June 1986 was estimated by the skippers to be different to that arising from the source data, it was assumed that the proportion of catch taken before the survey compared to the whole month was the same between the skippers' estimates and the source data. Thus length frequencies for 1-19 May 1985 and 1-9 June 1986 were calculated as follows:  $N_{l,partmonth} = N_{l,fullmonth} \times \text{SkipperTonnage}_{partmonth} / \text{SkipperTonnage}_{fullmonth}$  using the data in the below table.

|           | Days for which catch is required | Catch for the month (tons) | Skipper estimated catch for the month (tons) | Skipper estimated catch prior to the survey (tons) |
|-----------|----------------------------------|----------------------------|----------------------------------------------|----------------------------------------------------|
| May 1985  | 1-19 <sup>th</sup>               | 74245                      | 77174                                        | 48396                                              |
| June 1986 | 1-9 <sup>th</sup>                | 64662                      | 68189                                        | 10338                                              |

The LFs of catch from 1 April to the day before the survey are also stored in *AnchovyCatchLength.xlsx*, together with the corresponding observed tonnage. The "raised" length frequency used in the assessment is calculated using the same method as described above. The number of recruits landed between 1 April and the day before the commencement of the survey is calculated within the model from these "raised" length frequencies using the estimated selectivity-at-age 0 (equation A.17 of de Moor 2024).

#### **November Survey Data**

The time series of total biomass estimates and associated CVs from the acoustic surveys in November each year, corresponding to the standard survey area between Hondeklip Bay and Port Alfred, are given in Table 2. Daily egg production method (DEPM) estimates of anchovy adult biomass between 1984 and 1993 are also available (Table 2). For assessment purposes we assume this corresponds to spawning biomass. These data are stored in *SurveyData.xls*. The early part of this time series was updated by de Moor et al. (2008) so that the full time series corresponds to uncapped survey biomass determined with a new target strength formula (Coetzee et al. 2008). Coetzee et al. (2024a) recently revised the full time series to standardise strata across years, to extrapolate the biomass to the full survey area in years for which the survey did not cover the full area and to include the new length-weight regression  $w_l = 0.006803 \times l^{3.11124}$  (de Moor 2023). Note that the regression used to convert capped to uncapped data has not been updated from that used by de Moor et al. (2008) with the revised and extended timeseries.

Length frequencies are also available for the hydroacoustic surveys and are stored in *AnchovySurveyLength.xlsx*.

A survey in February/March 2024 was conducted after the November 2023 survey was cancelled (Coetzee et al. 2024b). The assessment fits model predicted biomass and associated proportions-at-length from 15<sup>th</sup> February 2024 (rather than 15<sup>th</sup> November 2023) to these February/March 2024 data (see de Moor 2024 for further details).

### Recruit Survey Data

The time series of recruit biomass and associated CVs from the May/June recruit surveys is given in Table 3. The recruit numbers at the time of the survey were calculated by summing the number of fish smaller than a cut-off length in the weighted length frequency (Appendix B<sup>1</sup>). In the assessments, the recruit numbers are used together with the CVs on recruit biomass. These survey data are stored in *SurveyData.xls*. The early part of this time series was updated by de Moor et al. (2008) and revised again in 2011 so that the full time series corresponds to uncapped recruitment determined with a new target strength formula (Coetzee et al. 2008). Coetzee et al. (2024a) recently revised the full time series to standardise strata across years, to extrapolate the biomass and recruit numbers to the full survey area in years for which the survey did not cover the full area and to include the new length-weight regression  $w_l = 0.006803 \times l^{3.11124}$  (de Moor 2023). Note that the regression used to convert capped to uncapped data has not been updated from that used by de Moor et al. (2008) with the revised timeseries.

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<sup>1</sup> An alternative method of deriving the WLF (converting the acoustic biomass to numbers at length) has been implemented for sardine in November surveys. This does away with the weighting of each trawl LF by the acoustic density assigned to it and rather uses a L/W regression to calculate numbers at length based on proportion at length from the pooled stratum sample LFs (Coetzee and Merkle 2011). This adjustment has not yet been implemented for anchovy in either the November or recruit surveys and will result in revised number of recruits and recruit mean weight. The extent of such change is unknown at this stage.

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**Table 1.** The monthly anchovy commercial catch tonnage (in thousands of tons) landed.

| Year | Jan    | Feb     | Mar     | Apr    | May    | Jun    | Jul    | Aug    | Sep    | Oct    | Nov   | Dec   |
|------|--------|---------|---------|--------|--------|--------|--------|--------|--------|--------|-------|-------|
| 1984 | 4.743  | 16.698  | 30.497  | 49.975 | 69.031 | 53.193 | 18.193 | 0.000  | 0.000  | 22.878 | 7.281 | 0.000 |
| 1985 | 2.284  | 11.296  | 13.005  | 65.782 | 74.245 | 83.354 | 20.954 | 0.938  | 0.784  | 0.000  | 0.000 | 0.000 |
| 1986 | 0.694  | 13.890  | 30.537  | 69.041 | 47.203 | 64.662 | 37.707 | 28.787 | 7.068  | 0.000  | 0.000 | 0.000 |
| 1987 | 61.838 | 114.134 | 139.674 | 48.084 | 14.925 | 50.851 | 78.632 | 67.886 | 24.357 | 0.000  | 0.000 | 0.000 |
| 1988 | 23.643 | 108.117 | 60.589  | 11.366 | 50.259 | 74.418 | 60.868 | 70.459 | 38.844 | 71.175 | 3.008 | 0.000 |
| 1989 | 10.582 | 62.255  | 24.864  | 59.553 | 83.796 | 39.623 | 13.747 | 0.000  | 0.000  | 0.000  | 0.000 | 0.000 |
| 1990 | 1.485  | 0.649   | 13.561  | 38.471 | 37.069 | 59.514 | 0.559  | 0.216  | 0.035  | 0.000  | 0.000 | 0.000 |
| 1991 | 0.502  | 2.923   | 29.246  | 36.466 | 22.908 | 51.769 | 6.106  | 1.091  | 0.025  | 0.009  | 0.000 | 0.000 |
| 1992 | 28.370 | 11.826  | 13.471  | 68.665 | 58.777 | 35.221 | 44.465 | 57.044 | 26.366 | 4.836  | 0.002 | 0.002 |
| 1993 | 18.851 | 20.643  | 17.703  | 43.292 | 14.102 | 1.196  | 10.878 | 67.182 | 38.890 | 3.052  | 0.015 | 0.065 |
| 1994 | 2.156  | 19.528  | 22.211  | 19.194 | 42.200 | 17.431 | 0.279  | 30.109 | 2.830  | 0.003  | 0.000 | 0.000 |
| 1995 | 0.089  | 0.406   | 20.440  | 22.748 | 13.473 | 35.621 | 32.335 | 39.432 | 1.873  | 10.396 | 1.617 | 0.000 |
| 1996 | 3.737  | 5.570   | 2.779   | 3.853  | 10.078 | 14.578 | 0.242  | 0.001  | 0.002  | 0.021  | 0.019 | 0.000 |
| 1997 | 0.026  | 0.014   | 0.000   | 0.022  | 1.196  | 0.758  | 20.434 | 10.927 | 23.355 | 3.653  | 0.000 | 0.000 |
| 1998 | 0.005  | 0.691   | 1.132   | 18.537 | 23.156 | 42.516 | 12.263 | 3.776  | 4.407  | 1.383  | 0.000 | 0.000 |
| 1999 | 0.000  | 0.000   | 3.548   | 8.490  | 19.660 | 28.585 | 20.120 | 33.472 | 51.708 | 13.273 | 1.037 | 0.000 |
| 2000 | 0.000  | 0.026   | 32.352  | 30.594 | 41.620 | 16.153 | 50.980 | 55.191 | 34.363 | 5.196  | 0.795 | 0.022 |
| 2001 | 0.949  | 8.547   | 8.285   | 34.781 | 34.083 | 45.297 | 10.296 | 30.812 | 51.150 | 60.380 | 2.735 | 0.195 |
| 2002 | 1.030  | 6.296   | 0.179   | 22.147 | 6.965  | 48.724 | 48.338 | 35.085 | 44.024 | 0.048  | 0.302 | 0.308 |
| 2003 | 0.019  | 0.041   | 3.524   | 16.636 | 23.991 | 78.245 | 48.150 | 16.801 | 40.602 | 27.159 | 3.479 | 0.232 |
| 2004 | 0.215  | 1.313   | 1.447   | 18.697 | 39.884 | 20.901 | 65.994 | 23.176 | 16.266 | 0.734  | 1.249 | 0.219 |
| 2005 | 0.000  | 2.018   | 27.919  | 51.448 | 56.259 | 21.245 | 42.187 | 27.097 | 43.257 | 9.318  | 1.951 | 0.028 |
| 2006 | 0.670  | 2.317   | 3.459   | 6.943  | 7.536  | 31.404 | 35.579 | 20.658 | 22.852 | 2.182  | 0.492 | 0.093 |
| 2007 | 0.009  | 0.145   | 2.252   | 17.700 | 58.403 | 31.054 | 34.540 | 37.774 | 43.707 | 25.136 | 0.681 | 1.691 |
| 2008 | 0.099  | 2.649   | 9.632   | 23.948 | 34.912 | 21.353 | 26.454 | 59.198 | 28.840 | 49.958 | 8.826 | 0.779 |
| 2009 | 1.747  | 11.461  | 9.782   | 13.018 | 18.797 | 8.903  | 38.577 | 35.513 | 27.714 | 6.081  | 2.873 | 0.000 |
| 2010 | 0.962  | 6.405   | 17.317  | 26.661 | 15.218 | 39.444 | 65.833 | 40.072 | 4.962  | 0.079  | 0.110 | 0.000 |
| 2011 | 0.037  | 3.177   | 3.136   | 16.316 | 24.467 | 17.060 | 40.931 | 14.733 | 0.016  | 0.005  | 0.000 | 0.000 |
| 2012 | 8.795  | 19.050  | 43.246  | 36.001 | 57.935 | 32.047 | 43.947 | 21.848 | 31.945 | 10.669 | 1.820 | 0.000 |
| 2013 | 3.268  | 0.073   | 0.025   | 23.381 | 17.990 | 12.717 | 3.384  | 3.530  | 7.995  | 3.132  | 3.295 | 0.012 |
| 2014 | 0.030  | 26.265  | 45.775  | 48.831 | 43.119 | 3.936  | 21.955 | 19.855 | 21.962 | 7.592  | 1.167 | 0.012 |
| 2015 | 0.015  | 4.478   | 47.205  | 54.216 | 68.953 | 28.559 | 29.077 | 4.308  | 0.224  | 0.138  | 0.937 | 0.018 |
| 2016 | 0.063  | 5.447   | 5.141   | 51.354 | 44.594 | 64.863 | 41.486 | 22.310 | 22.861 | 3.388  | 0.001 | 0.000 |
| 2017 | 0.046  | 0.170   | 4.448   | 47.149 | 46.408 | 11.363 | 76.731 | 22.675 | 4.997  | 1.169  | 0.063 | 1.125 |
| 2018 | 0.002  | 16.344  | 28.523  | 33.738 | 48.608 | 12.944 | 39.138 | 61.431 | 8.427  | 3.348  | 0.744 | 0.119 |
| 2019 | 0.125  | 6.633   | 17.751  | 23.436 | 33.308 | 20.625 | 18.990 | 15.273 | 27.867 | 0.775  | 0.006 | 0.009 |
| 2020 | 0.096  | 8.364   | 26.852  | 68.041 | 59.137 | 11.949 | 39.938 | 20.972 | 39.810 | 9.889  | 0.132 | 0.000 |
| 2021 | 3.643  | 2.440   | 2.285   | 22.473 | 37.992 | 38.649 | 12.813 | 27.539 | 6.806  | 0.469  | 1.127 | 0.000 |
| 2022 | 0.000  | 0.757   | 4.346   | 40.873 | 57.077 | 43.639 | 8.037  | 8.138  | 8.963  | 0.037  | 0.524 | 0.019 |
| 2023 | 0.005  | 0.874   | 1.820   | 31.701 | 66.179 | 6.774  | 0.140  | 0.420  | 0.320  | 0.515  | 0.161 | 0.002 |

**Table 2.** Anchovy total biomass (in tons) west of Port Alfred and associated CV from the November hydroacoustic survey and anchovy spawner biomass (in tons) and associated CV determined by the DEPM.

| Year     | Hydroacoustic survey |       | DEPM            |       |
|----------|----------------------|-------|-----------------|-------|
|          | Total Biomass        | CV    | Spawner Biomass | CV    |
| 1984     | 1 340658             | 0.285 | 1100000         | 0.450 |
| 1985     | 1 344637             | 0.170 | 616000          | 0.400 |
| 1986     | 2 582983             | 0.171 | 2001000         | 0.350 |
| 1987     | 2 127850             | 0.157 | 1606000         | 0.300 |
| 1988     | 1 622152             | 0.222 | 1679000         | 0.350 |
| 1989     | 751529               | 0.167 | 421000          | 0.350 |
| 1990     | 651735               | 0.183 | 723000          | 0.580 |
| 1991     | 2 353743             | 0.162 | 2913000         | 0.350 |
| 1992     | 2 098563             | 0.163 | 3600000         | 0.310 |
| 1993     | 916359               | 0.209 | 770000          | 0.340 |
| 1994     | 617555               | 0.159 |                 |       |
| 1995     | 601271               | 0.217 |                 |       |
| 1996     | 162048               | 0.410 |                 |       |
| 1997     | 1 482633             | 0.267 |                 |       |
| 1998     | 1 229132             | 0.217 |                 |       |
| 1999     | 2 052156             | 0.156 |                 |       |
| 2000     | 4 714696             | 0.127 |                 |       |
| 2001     | 6 720287             | 0.107 |                 |       |
| 2002     | 3 882747             | 0.154 |                 |       |
| 2003     | 3 563232             | 0.236 |                 |       |
| 2004     | 2 037011             | 0.131 |                 |       |
| 2005     | 3 077001             | 0.144 |                 |       |
| 2006     | 2 106273             | 0.136 |                 |       |
| 2007     | 2 505655             | 0.157 |                 |       |
| 2008     | 3 598790             | 0.120 |                 |       |
| 2009     | 3 792547             | 0.136 |                 |       |
| 2010     | 2 058185             | 0.144 |                 |       |
| 2011     | 754124               | 0.204 |                 |       |
| 2012     | 3 187964             | 0.116 |                 |       |
| 2013     | 3 819666             | 0.102 |                 |       |
| 2014     | 2 970760             | 0.137 |                 |       |
| 2015     | 1 944258             | 0.157 |                 |       |
| 2016     | 1 730917             | 0.228 |                 |       |
| 2017     | 1 568398             | 0.196 |                 |       |
| 2018     | 1 559546             | 0.235 |                 |       |
| 2019     | 850517               | 0.206 |                 |       |
| 2020     | 2 556598             | 0.164 |                 |       |
| 2021     | -                    | -     |                 |       |
| 2022     | 1 021069             | 0.220 |                 |       |
| Feb 2024 | 1 013827             | 0.485 |                 |       |

**Table 3.** The date of the commencement of the annual recruit survey (taken to be the date on which the first acoustic interval is logged and not the date on which the first station was occupied) and the cut-off lengths used to estimate juvenile anchovy from the recruit surveys, anchovy recruitment (in tons and numbers in billions) from Hondeklip Bay to Cape Infanta and associated CV from the recruitment acoustic survey, and the mean recruit weight (in grams).

| Year | Date of commencement of survey | Time of the recruit survey after 1 May | Cut-off length (cm) for anchovy recruits in the survey | Biomass  | CV*   | Mean Weight | Numbers* |
|------|--------------------------------|----------------------------------------|--------------------------------------------------------|----------|-------|-------------|----------|
| 1985 | 20-May                         | 0.613                                  | <8.0                                                   | 117234   | 0.403 | 2.36        | 49.767   |
| 1986 | 10-Jun                         | 1.300                                  | <9.0                                                   | 521313   | 0.175 | 4.05        | 128.717  |
| 1987 | 20-Jul                         | 2.613                                  | <7.5                                                   | 107804   | 0.399 | 2.67        | 40.412   |
| 1988 | 27-Jun <sup>2</sup>            | 1.867                                  | <7.0                                                   | 101985   | 0.211 | 2.13        | 47.777   |
| 1989 | 08-Jun <sup>3</sup>            | 1.233                                  | <7.0                                                   | 25917    | 0.479 | 2.10        | 12.345   |
| 1990 | 22-Jun                         | 1.700                                  | <8.5                                                   | 133232   | 0.269 | 2.92        | 45.578   |
| 1991 | 07-May                         | 0.194                                  | <8.5                                                   | 239361   | 0.203 | 3.34        | 71.586   |
| 1992 | 13-May                         | 0.387                                  | <8.5                                                   | 165582   | 0.200 | 2.97        | 55.705   |
| 1993 | 21-May                         | 0.645                                  | <8.5                                                   | 139006   | 0.207 | 1.88        | 73.936   |
| 1994 | 05-May                         | 0.129                                  | <8.5                                                   | 79797    | 0.198 | 3.67        | 21.741   |
| 1995 | 10-Jun                         | 1.300                                  | <9.5                                                   | 353262   | 0.171 | 3.34        | 105.822  |
| 1996 | 05-Jun                         | 1.133                                  | <8.5                                                   | 51658    | 0.248 | 2.31        | 22.321   |
| 1997 | 17-May                         | 0.516                                  | <9.0                                                   | 305468   | 0.187 | 3.99        | 76.502   |
| 1998 | 20-May                         | 0.613                                  | <9.5                                                   | 381715   | 0.156 | 2.97        | 128.707  |
| 1999 | 10-May                         | 0.290                                  | <9.0                                                   | 625411   | 0.186 | 3.67        | 170.254  |
| 2000 | 15-May                         | 0.452                                  | <9.5                                                   | 2 465941 | 0.168 | 3.96        | 623.012  |
| 2001 | 05-May                         | 0.129                                  | <8.5                                                   | 1 769930 | 0.139 | 2.96        | 598.470  |
| 2002 | 05-May                         | 0.129                                  | <9.5                                                   | 1 206766 | 0.124 | 2.51        | 479.850  |
| 2003 | 14-May                         | 0.419                                  | <10.0                                                  | 1 396638 | 0.189 | 3.25        | 430.308  |
| 2004 | 08-May                         | 0.226                                  | <9.5                                                   | 729333   | 0.216 | 3.62        | 201.737  |
| 2005 | 13-May                         | 0.387                                  | <8.0                                                   | 253228   | 0.223 | 2.00        | 126.691  |
| 2006 | 19-May                         | 0.581                                  | <9.0                                                   | 242583   | 0.159 | 2.12        | 114.480  |
| 2007 | 18 May                         | 0.548                                  | <9.0                                                   | 1 309951 | 0.197 | 2.73        | 480.065  |
| 2008 | 21 May                         | 0.645                                  | <10.0                                                  | 1 507693 | 0.202 | 2.63        | 573.429  |
| 2009 | 15 May                         | 0.452                                  | <9.0                                                   | 844721   | 0.231 | 2.77        | 304.407  |
| 2010 | 27 May                         | 0.839                                  | <8.5                                                   | 732518   | 0.302 | 2.83        | 259.092  |
| 2011 | 27 May                         | 0.839                                  | <8.5                                                   | 246548   | 0.284 | 2.51        | 98.145   |
| 2012 | 16 June                        | 1.500                                  | <8.0                                                   | 246999   | 0.231 | 2.99        | 82.703   |
| 2013 | 23 May                         | 0.710                                  | <8.5                                                   | 803100   | 0.195 | 2.76        | 291.314  |
| 2014 | 10 May                         | 0.290                                  | <9.5                                                   | 595586   | 0.336 | 3.18        | 187.265  |
| 2015 | 22 May                         | 0.677                                  | <8.5                                                   | 588802   | 0.181 | 3.21        | 183.210  |
| 2016 | 8 Jun                          | 1.233                                  | <8.5                                                   | 253890   | 0.179 | 3.33        | 76.295   |
| 2017 | 12 Jun                         | 1.367                                  | <8.5                                                   | 2 211337 | 0.143 | 2.89        | 766.470  |
| 2018 | -                              | -                                      | -                                                      | -        | -     | -           | -        |
| 2019 | 7 Jun <sup>4</sup>             | 1.200                                  | <8.5                                                   | 507412   | 0.282 | 1.98        | 256.205  |
| 2020 | 17 Jun                         | 1.533                                  | <8.5                                                   | 1 464887 | 0.187 | 3.24        | 451.444  |
| 2021 | 9 Jun                          | 1.267                                  | <8.5                                                   | 386508   | 0.170 | 2.66        | 145.409  |
| 2022 | 2 Jul                          | 2.032                                  | <8.0                                                   | 246201   | 0.201 | 1.51        | 162.933  |
| 2023 | 25 May                         | 0.774                                  | <9.0                                                   | 54590    | 0.226 | 3.31        | 16.500   |

<sup>2</sup> The first station was on 27<sup>th</sup> June 1988, although the first acoustic interval was only logged after midnight, i.e. on 28<sup>th</sup> June 1988.

<sup>3</sup> The first station was on 8<sup>th</sup> June 1989, although the first acoustic interval was only logged after midnight, i.e. on 9<sup>th</sup> June 1989.

<sup>4</sup> The first station was on the 5<sup>th</sup> June 2019, although the first acoustic interval was logged on the 7<sup>th</sup> June 2015.

## Appendix A: Pelagic sample allocation

The sample allocation method is the process whereby a length frequency (LF) is allocated to every commercial landing, enabling the transformation of the catch to its raised length frequency (RLF). The commercial catch data and field station length frequency data are entered and stored on a Sybase database on the DFFE network. Length frequency data from the Fisheries observer program (2001 to 2011), SAPFIA-funded observer program (2017-presently ongoing) and Industry canneries (2011 to 2018 and from 2020 and presently ongoing) are included in the calculation. The calculations are performed in Access.

### Species

It is necessary to calculate LFs for conditioning the assessments for anchovy (*Engraulis encrasicolus*), sardine (*Sardinops sagax*) and round herring (*Etrumeus whiteheadii*); LFs for horse mackerel (*Trachurus trachurus capensis*) are also generated for every run.

### Data sources

- Commercial catch: The skipper completes a Pelagic catch Report for every trip and records the estimated catch per set. From 1984 to 2010 this was recorded by a pelagic 10' by 10' block, while from 2011 skippers provided actual latitude and longitudes for each throw as well as the block number position. Every landing is sampled for its species composition and weighed. The fisheries inspectors/scale monitors complete the Pelagic Landing Sheet, based on the actual tonnage offloaded at the landing site. Skipper data (Pelagic Catch Reports) are available on Sybase from 1984 onwards but inspector data (Pelagic landing Sheets) were obtained only from 1987. DFFE field station personnel collect data sheets and enter the information on Sybase.
- Field station samples: DFFE field station personnel collect random samples at the major pelagic fishing harbours for species composition and length frequency (Capricorn Fishing Monitoring was contracted from 2002 until 2005 to man St. Helena Bay and Gansbaai). Samples of industrial fish such as anchovy and round herring are obtained from the top of the hold before the vessel discharges. For this reason industrial samples are obtained mainly from the last throw of the trip. Offloading further damages the already partially-decomposed fish and one cannot sample from the conveyer belt because it would be impossible to weigh those fish. Directed sardine catch, on the other hand, is kept in a very good condition onboard on ice and good quality samples are easily obtained from the conveyor belt, whilst the vessel is discharging. Unfortunately, it is seldom possible to establish which throw is being sampled. Field station data are available on Sybase from 1984 onwards. Ports sampled over the period include Lamberts Bay, Laaiplek, St. Helena Bay, Saldanha, Cape Town, Hout Bay, Kalk Bay, Hermanus, Gansbaai, Mossel Bay and Port Elizabeth. In 2017 it was agreed that a standard seawater allowance of 4.5% for anchovy and round herring and 2.6% for sardine would be implemented (DFFE 2017); this has been implemented since the beginning of the 2018 season.
- Observer samples: The observer program started in 1999 but was terminated in 2011. An industry-funded observer program began in 2017 and is ongoing. Observer samples from 2017 to 2024 have been incorporated into the data thus far. Onboard biological sampling by observers was started in 2001. Observer sampling results reflect an improvement on the field station data because samples are obtained from a known throw, all throws are sampled and the fish is always in a good condition. Unfortunately, the length frequency samples have to be taken ashore for weighing and this gives rise to room for error. The data are stored in an Access database called CAPFISH.

- Industry cannery samples: As part of the quality process at the canneries, fish are also measured and weighed according to DFFE field station specification.

#### Data extraction from Sybase

- Catch data are extracted from Sybase as text (flat) files; *throw.csv* contains the skippers' data (Pelagic Catch Report) which is proportioned to the actual landed mass and *catch.csv* contains the inspectors' data (Pelagic Landing Sheet).
- Field station data are extracted in the same manner; *spcomp.csv* contains the species composition data and *lfreq.csv* contains the length frequency data.

#### Data handling and evaluation

##### DFFE data

- Unfortunately, there is no manual proof reading of all the data, except in cases where the number of throws is excessive (more than 10) and the trip duration is of an unrealistic duration (more than 3 days). Data validation is limited to electronic checking for noticeable mistakes.
- A duplicate dataset of *catch.csv* which is regularly updated by email is kept at Saldanha in an Access table. This means that the data are entered twice, but into separate databases and this allows for the comparison of the two data sets on a regular basis for differences and errors. It might appear unnecessary to keep two data sets, but this is the sole reason that the pelagic catch data remain representative of what was recorded by the scale monitors. The entering of data into Access was stopped since 2015 because these data are now obtained from Industry and used to verify the catches.
- The expected sample weights associated with the length frequency data in *lfreq.csv* are computed and samples that deviate more than 30% are flagged and checked against the raw data. If a flag results from a punch error then the data are corrected, but in the case of a sampling error the record is deleted from the data base.
- Suspect positions, for example areas outside the normal catch areas are checked against the raw data and, if necessary, corrected.

##### Observer data

- Limited manual proof reading of data.
- Only observer trips that match the commercial data for vessel name and date are used. Mismatched dates do occur, making it very difficult to establish whether a specific vessel carried an observer on a specific date. Therefore, samples from such observer trips are ignored to prevent the inclusion of poor data. Only trips that do link can be used, because the scale monitor's species composition is used to determine the target species of the length frequency sample.
- The structure of the observer length frequency table is altered to make it compatible with the Sybase dataset.
- Only observer length frequencies whose predicted sample weights fall within the set range are used. Data with possible measurement errors or wrong species names are excluded.

#### Access programs

RLFdata.accdb (where the LFs are generated)

This program has links to the following data sets that are required for the computation.

- 1) Cannery data.accdb (sardine length frequency)
- 2) RSAP reference.accdb (data set with vessel detail and pool positions)
- 3) Capfish.accdb (if data from 2001 to 2011 and from 2017 onwards are included)
- 4) Pel Catch (skipper data sets)

### General program outline

- Catches are allocated to pool-area/week strata:
  1. Week: the throw date with the largest catch is used.
  2. Pool area: the existing 21 areas (see Figure A.2) are used, but from 1999 onwards area 21 was subdivided into areas 23 and 24, to accommodate the eastward fishing expansion. The throws within each landing are examined, and the throw with the greatest mass is used as the representative throw.
  3. Assign a target species to every catch. The species with the largest mass is defined as the dominant species in the landing.
- The length frequency samples are grouped by species and target species for the pool-area/week strata and summed.
- A new catch table with additional space for the allocated length frequencies is created.
- The length frequency table is searched and a frequency based on the species, target species, week and pool area criteria are assigned to the catch table.
- In the event of catches not being represented by an appropriate sample, the pool-area/week will be expanded to include surrounding areas and weeks. Stratum expansion continues alternately by week and pool until an appropriate frequency is located.
- If no appropriate sample is found then the average sample for the month is applied. Where no sample for the month exists, the length frequency is estimated using the length frequency of a former month as detailed in the text. Catches of each species and the length frequencies are summed by month over larger user specified areas.
- The length frequencies are exported as Excel files in numbers per length group.

The user specified areas that are used are:

- A. Areas 1-6: North of Cape Columbine
- B. Areas 7-12: Cape Columbine to Cape Point
- C. Areas 13-20: Cape Point to Cape Infanta
- D. Area 23: Cape Infanta to Plettenberg Bay
- E. Area 24: East of Plettenberg Bay

In 2007, the border between area 23 and 24 was shifted slightly west to 24 degrees east (Tsitsikama), although this made little difference in practice since catches between 23 and 24 were small.

Although the LFs are summarized according to different areas, the allocation process is still based on the original pool areas, with the exception of those cases where pool areas were split by the new borders.

## Program changes

In January 2007 four changes were made to the process above:

- The observer length frequencies were included.
- To prevent juvenile sardine frequencies from being allocated to adult sardine catches, the species was separated into directed and bycatch for allocation purposes. This is applicable only when sardine is landed as a bycatch with anchovy. Sardine bycatch with anchovy is mainly juvenile fish whereas bycatch with round herring it is mostly adult fish.
- Noticeable errors in the LF results when the field station catch composition data are used to identify the target species of the length frequency sample, and these composition data differ from those of the scale monitor. Because the field station data are not proofread, and given the inclusion of the observer length frequencies (they also need a target species to be identified), it was decided to standardize on the scale monitors species composition as the only source.
- Missing skipper data (catch area) are catered for. This occurs when the skipper fails to hand in a trip sheet. Currently this is not a major problem but it did happen in the 1980s and 1990s. Where the *catch.csv* file does not have a related record in the *throw.csv* file, the program will search for the most likely catch position, based on the catch type of the other vessels for the same date.

From 2012 the cannery length frequencies are included.

The first change leads to enhanced coverage, especially in the case of industrial fish, i.e. anchovy that are poorly sampled by the field stations. The last three changes were implemented to prevent errors caused by bad data or poor sampling coverage. This can typically be seen in a LF plot as an improbable peak at a certain length group.

In March 2007 an additional change was implemented. Towards the end of the year sporadic landings can be overlooked, because it is not cost effective to continue extensive sampling. These landings are generally small but it is still necessary to allocate a size to the fish. In the past the annual LF average was used, but it was decided to rather allocate the LF from the adjacent month. The LFs are first stratified by area and species type, but where no match is found the requirements for matching area and target species are removed alternatively until a match is found.

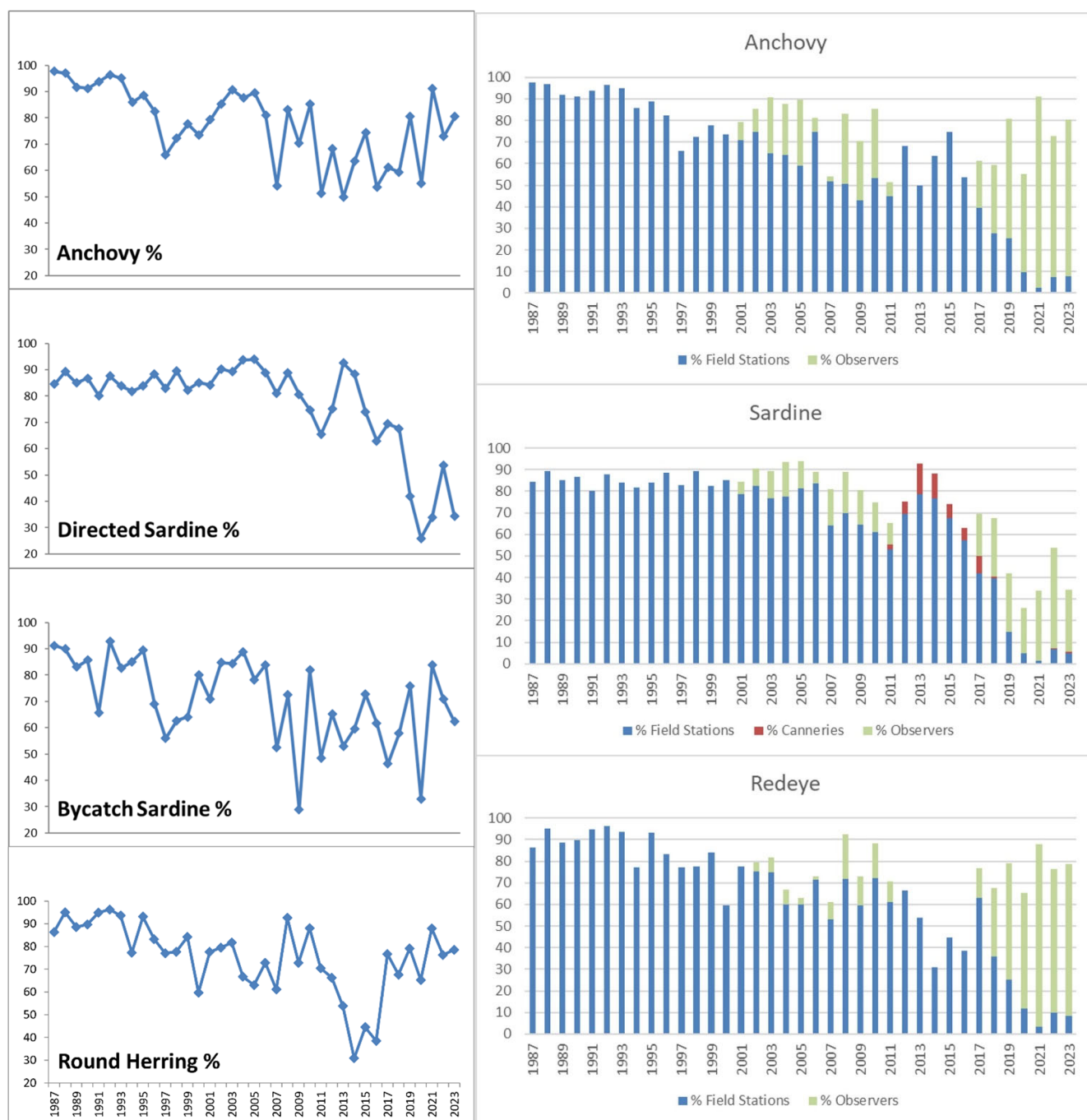
Even though throws in multiple pool areas during a single trip do occur, only the catch area for the biggest throw is selected. This is done in order to maintain continuity with the old sample allocation method. A change that could be considered would be to allocate a sample to every throw as opposed to every trip. The scale monitor samples at regular intervals and discrete throws are not sampled. However, if one assumes the species composition of the throws are uniform, then the catch per throw can be calculated, by proportionally applying the species composition to individual throws. Observer sampling is ideally suited for this approach, because every throw is sampled, but greater sampling coverage and matched skipper throws are required.

## Sampling coverage

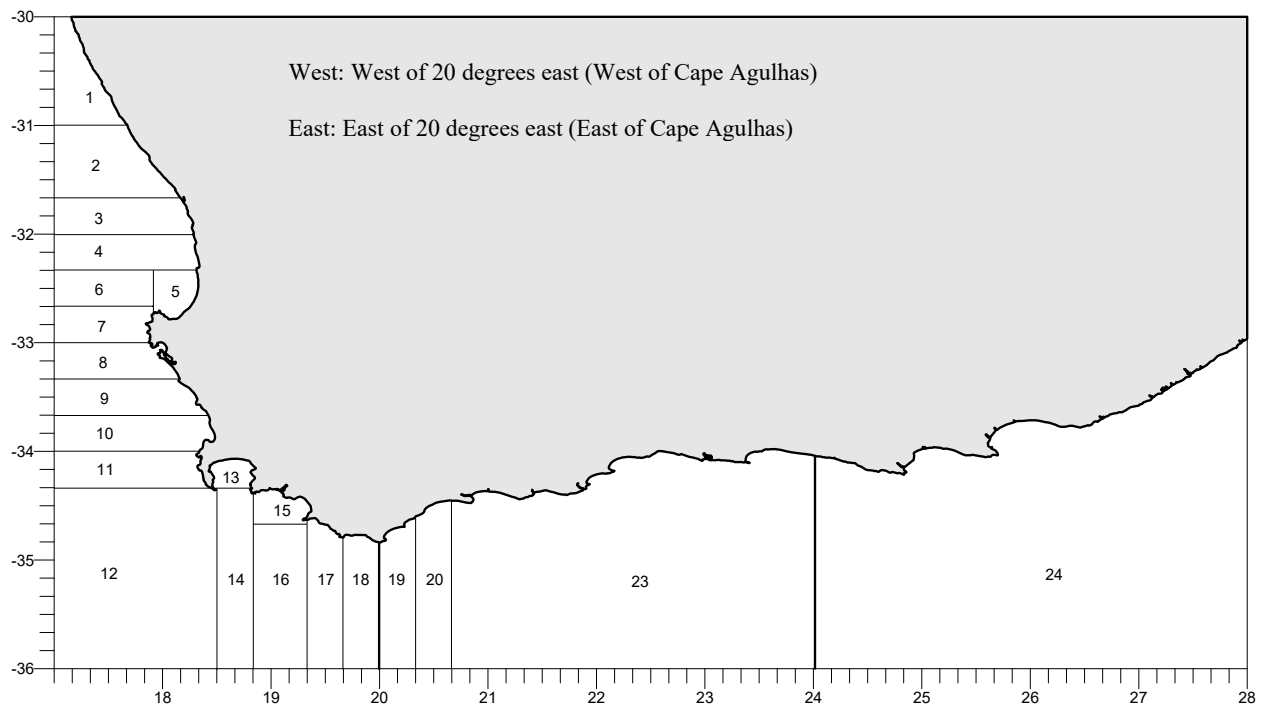
Optimum sample size and sampling coverage can be determined only by using a suitable statistical study, and one can therefore only speculate on the sample size required. Logistical constraints have necessitated a random stratified sampling method, and the grouping of catches and samples on a week/pool-area basis has been adopted since electronic data processing began. Both the sampling and the length frequency approaches are arguably the most suitable considering the fishing strategy and the available data. The percentage coverage per stratum is readily quantified, and the first level pool-area/week coverage could possibly be used as an index of sampling coverage. 100 percent coverage is not attainable because

of financial and logistical constraints, and it is more than likely unnecessary. From Figure A.1 it appears that 80 percent coverage is attainable.

Many factors influence the relationship between the number of samples taken and the coverage obtained, but in general more samples will lead to better coverage. In earlier years, field stations were well manned and more samples were taken than presently. The Observer program was introduced in 2000 and this improved sampling and as a result field station sampling was reduced. However, when the program was terminated in 2011 the field station coverage was not adjusted to earlier levels. This poor coverage can be seen as the low values on the graph in Figure A.1. The inclusion of the industry-funded observer samples since 2017 have assisted coverage in recent years (Coetzee and Merkle 2019). The fisheries cannery data (used since 2012) is necessary to maintain directed sardine sampling levels.



**Figure A.1.** Sampling coverage obtained on a first level pool-area/week including 2000-2011 and 2017 to 2023 observer data. The histograms indicate the proportion of sampling obtained from observer data.



**Figure A.2.** The pool areas that are used for sample allocation.

## Appendix B: Method Used to Calculate Recruit Biomass

To estimate recruit-only biomass and CVs (rather than that of recruits and adults combined) the acoustic density is apportioned between recruits and adults for each interval, based on the length- specific target strength and a cut-off length (derived independently from maturity at length information). Essentially, the density in each interval is multiplied by the proportion of recruits in that interval to get a recruit only density. The proportion of recruits in each interval is obtained by calculating the proportion of acoustic energy backscattered by recruits only, based on the length frequency that each interval has been assigned and a cut-off length:

- 1) For each trawl (grid) the acoustic back scattering for each length class is calculated for each species and multiplied by the number of fish in that length class (basically applying the species specific target strength relationship to the length class ( $L_t$  )):

$$BS = \begin{cases} 10^{0.1 \times -21.12} \times L_t^{-12.15/10} \times N & \text{if } Sp = 1 \\ 10^{0.1 \times -13.21} \times L_t^{-14.9/10} \times N & \text{if } Sp = 2 \text{ or } 5 \\ 10^{0.1 \times -7.75} \times L_t^{-15.44/10} \times N & \text{if } Sp = 3 \text{ or } 4 \end{cases}$$

where  $Sp\ 1$  = anchovy,  $Sp\ 2$  = sardine,  $Sp\ 3$  = horse mackerel,  $Sp\ 4$  = mackerel and  $Sp\ 5$  = round herring.

- 2) The backscattering ( $BS$ ) is summed for each species for each trawl to give a total backscatter for each grid,  $BS_{tot}$ .
- 3) The backscattering due to recruits,  $BS_{rec}$ , is then calculating by summing  $BS$  for only the length classes less than the cut-off length for each species for each trawl. The cut-off length is obtained from the length at 20% maturity, derived from samples collected during the survey (Coetzee et al. 2018, Yemane et al. 2024).
- 4) The proportion of recruits in each trawl is then calculated by  $BS_{rec}/BS_{tot}$ .
- 5) This proportion is then multiplied by the original interval density (of recruits and adults) to obtain the recruit only density per interval, which is then used to estimate the biomass and CV of recruits following the Jolly and Hampton (1990) method.
- 6) This recruit only density is used in the regressions<sup>5</sup> of capped to uncapped data in order to estimate (using the Jolly and Hampton weighting procedure) the uncapped recruit only biomass prior to 1997 together with a CV.

A weighted length frequency is then derived as follows:

- 7) Each acoustic interval is linked to a particular grid reference (trawl sample) which was used to scale the acoustic energy to density. The trawl sample has a length frequency (LF) and associated length frequency mass (LFMASS). This LF and LFMASS include both adults and recruits.
- 8) For each interval, the acoustic density is multiplied by the interval length. This weighted interval density is then summed over all intervals for each grid reference, per stratum and per species to give an acoustic weighting to each grid reference,  $w_{GR}(grid, stratum, species)$ .
- 9) The weighted grid reference is then summed over all grid references for each stratum and species to give a weighted grid reference per stratum for each species,  $w_{GR}(stratum, species)$ .
- 10) For each length class of each grid reference, a Trawl WF (trawl weighting factor) =  $w_{GR}(grid, stratum, species)/LFMASS$  is calculated. This converts the acoustic weighting (in terms of mass) into a factor in terms of numbers.
- 11) The length frequency (LF) is then weighted by this Trawl WF and summed for each length class to give a weighting to each length class (Lgroup) for each stratum for each species sum(number\*trawl WF),  $WLF(Lgroup, stratum, species)$ .

<sup>5</sup> The regressions were, however, calibrated with total (recruit and adult) densities (de Moor et al. 2008).

- 12)  $WLF(Lgroup, stratum, species)$  is then scaled to the biomass of the stratum:  $BLF(Lgroup, stratum, species) = [WLF(Lgroup, stratum, species) * Biomass(stratum, species)] / \sum w_{GR}(stratum, species)$ .
- 13) BLF is then summed across all strata for each species to give a final length frequency per species for the survey (this is done separately up to Cape Infanta and for the whole survey).